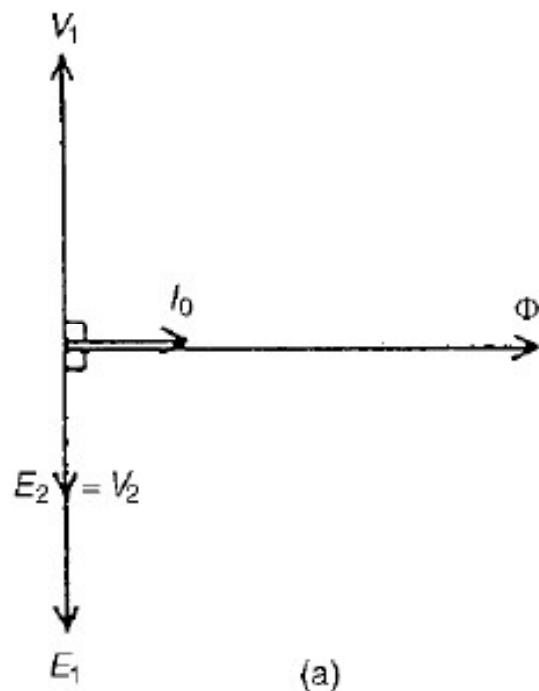


Lecture – 9 (Unit-3)

EE10510

Single-Phase Transformer

Phasor Diagram of Ideal Transformer



The core flux is common to both primary and secondary windings in a transformer and is thus taken as the reference phasor in a phasor diagram. On no-load the primary winding takes a small no-load current I_0 and since, with losses neglected, the primary winding is a pure inductor, this current lags the applied voltage V_1 by 90° . In

Ideal Single-Phase Transformer (Load Operation)

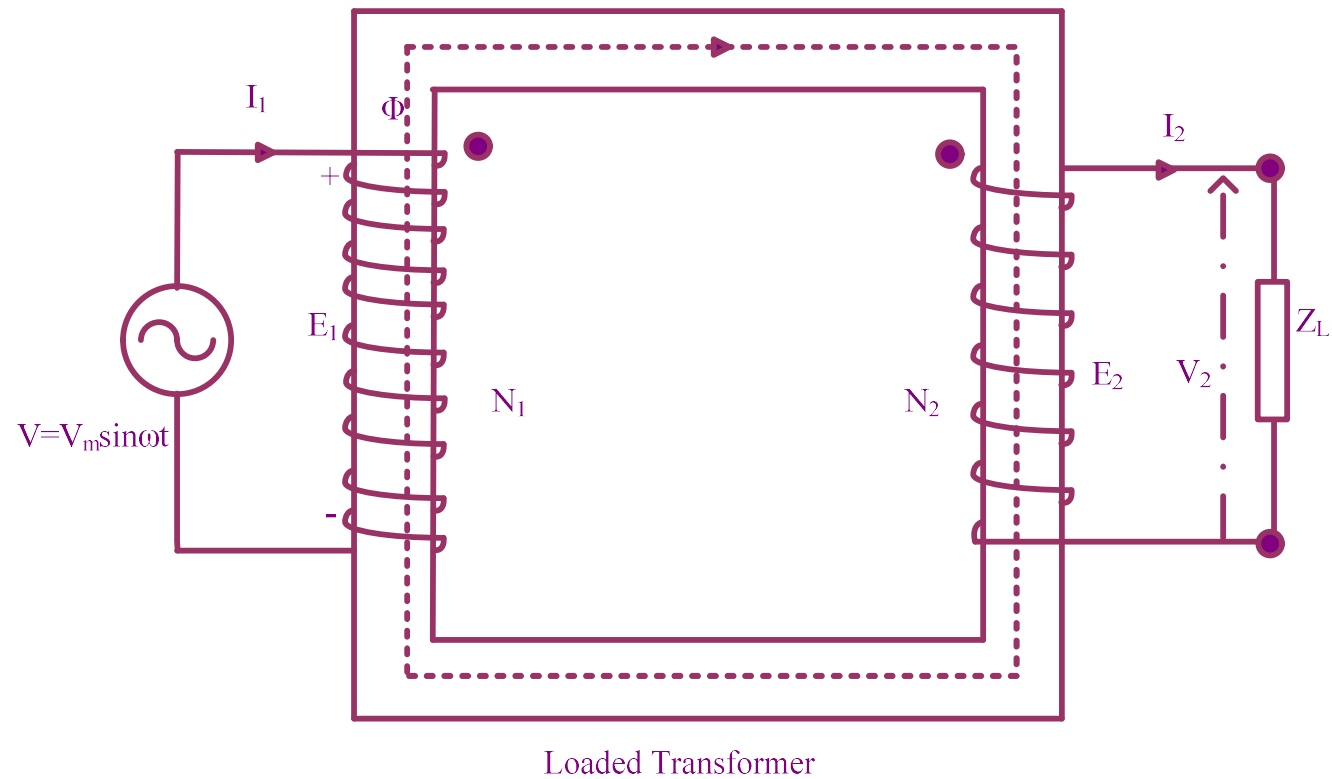


Fig. 2

- A load impedance Z_L is connected across the terminals of the secondary winding, results in load current flows as marked in Fig. 2.
- This load current produces a demagnetizing mmf and the flux tends to collapse. However this is detected by the primary immediately as both E_2 and E_1 tend to collapse.
- The current drawn from supply increases up to a point the flux in the core is restored back to its original value.
- The demagnetizing mmf produced by the secondary is neutralized by additional magnetizing mmf produces by the primary leaving the mmf and flux in the core as in the case of

no-load. Thus the transformer operates under constant induced emf mode. Thus $i_1 N_1 - i_2 N_2 = i_0 N_1$ but $i_0 = 0$

$$i_2 N_2 = i_1 N_1 \text{ and the rms value } I_2 N_2 = I_1 N_1.$$

The above equation can be written in phasor form as,

$$I_1 N_1 = I_2 N_2 \text{ or } I_1 = \frac{I_2 N_2}{N_1}$$

$$\text{Also } \frac{E_1}{E_2} = \frac{N_1}{N_2} = \frac{I_2}{I_1}$$

Thus voltage and current transformation ratio are inverse of one another. If an impedance of Z_L is connected across the secondary, then $I_2 = E_2 / Z_L$ or $Z_L = E_2 / I_2$

The input impedance under such conditions is

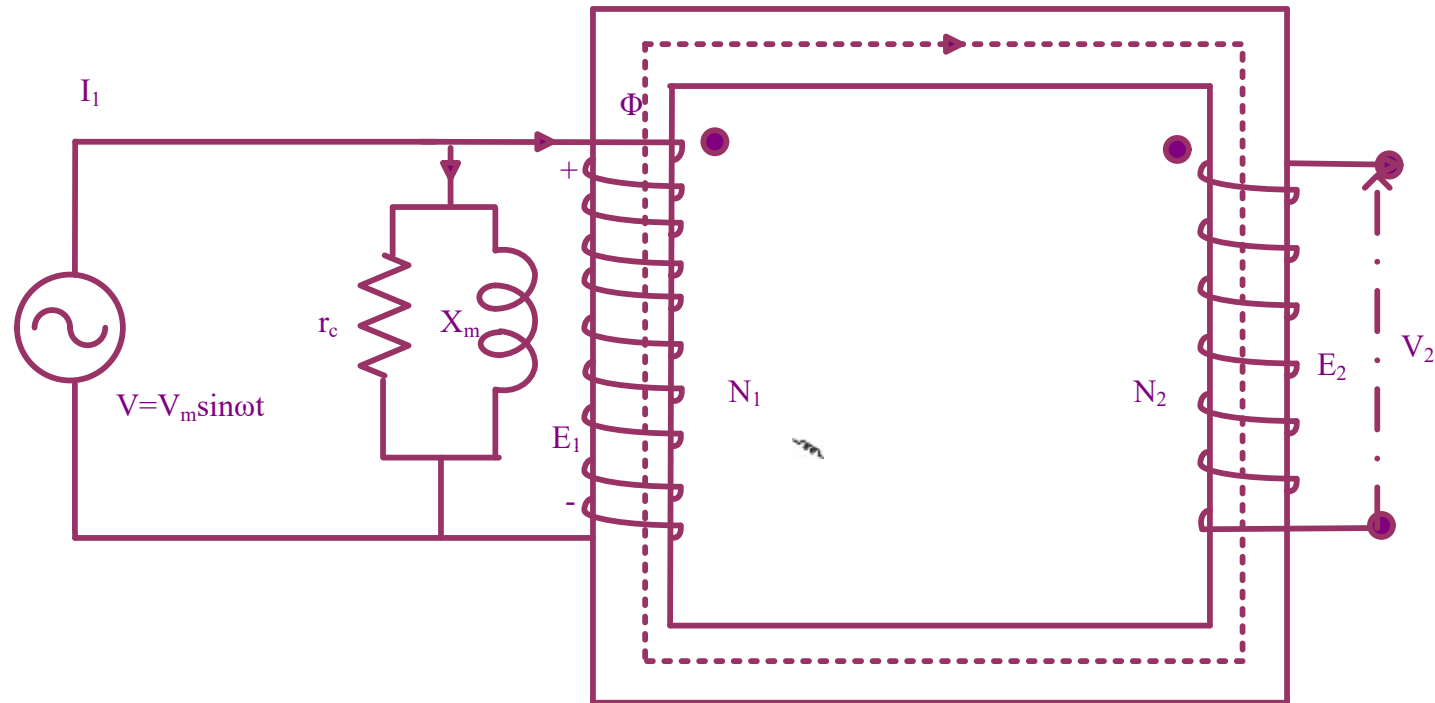
$$Z_i = \frac{E_1}{I_1} = \left(\frac{N_1}{N_2}\right)^2 \frac{E_2}{I_2} = \left(\frac{N_1}{N_2}\right)^2 Z_L$$

An impedance of Z_L when viewed ‘through’ a transformer of turns ratio (N_1/N_2) is seen as $(N_1/N_2)^2 \cdot Z_L$. Transformer thus acts as an impedance converter. The transformer can be interposed in between a source and a load to ‘match’ the impedance.

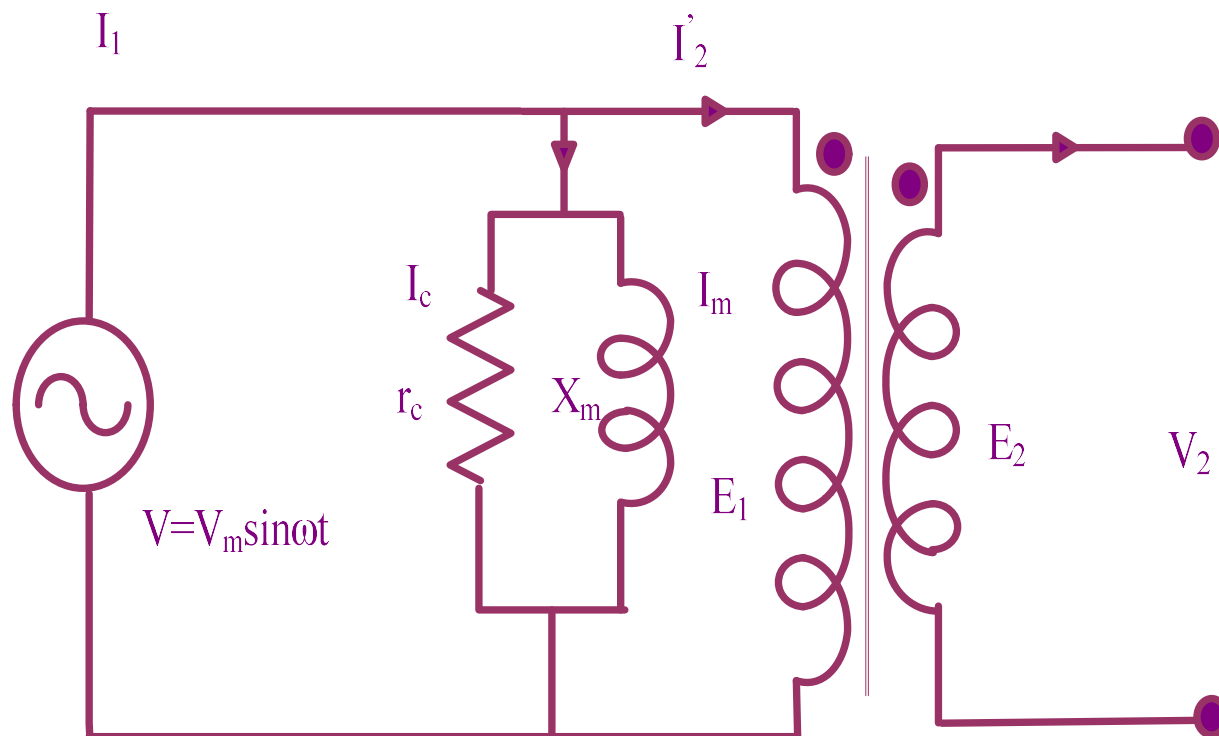
Practical Single-Phase Transformer

- An ideal transformer is useful in understanding the working of a transformer. But it cannot be used for the computation of the performance of a practical transformer due to the **non-ideal nature of the practical transformer**.
- In a working transformer the performance aspects like **magnetizing current, losses, voltage regulation, efficiency etc** are important. Hence the effects of the **non-idealization** like **finite permeability, saturation, hysteresis and winding resistances** have to be added to an ideal transformer to make it a practical transformer.

Practical Single-Phase Transformer



Physical Arrangement of Practical Transformer at no-Load



Equivalent Circuit of Practical Transformer at No-Load

- Finite permeability of the magnetic circuit necessitates a finite value of the current to be drawn from the mains to produce the mmf required to establish the necessary flux.
- The current and mmf required is proportional to the flux density B that is required to be established in the core.

$$B = \mu H; B = \Phi/A$$

- where A is the area of cross section of the iron core m^2 . H is the magnetizing force which is $H = Ni/l$ where l is the length of the magnetic path, m. or $\Phi = A\mu(Ni)/l = \text{Permeance} \times \text{mmf}$
- The magnetizing force and the current vary linearly with the applied voltage as long as the magnetic circuit is not saturated.

- For practical transformers core loss and iron loss exist.
- At no-load conditions, the primary winding is not fully reactive and it supplies, iron loss in the core and very small amount of copper loss in the primary.
- The no-load current lags behind V_1 by an angle Φ_0 which is less than 90° .
- The no-load input power is given by,

$$W_0 = V_1 I_0 \cos \Phi_0$$

Where $\cos \Phi_0$ is the no-load power factor under no-load conditions.

No-Load Primary
Current
 $I_0 = \text{Sqrt}(I_w^2 + I_\mu^2)$

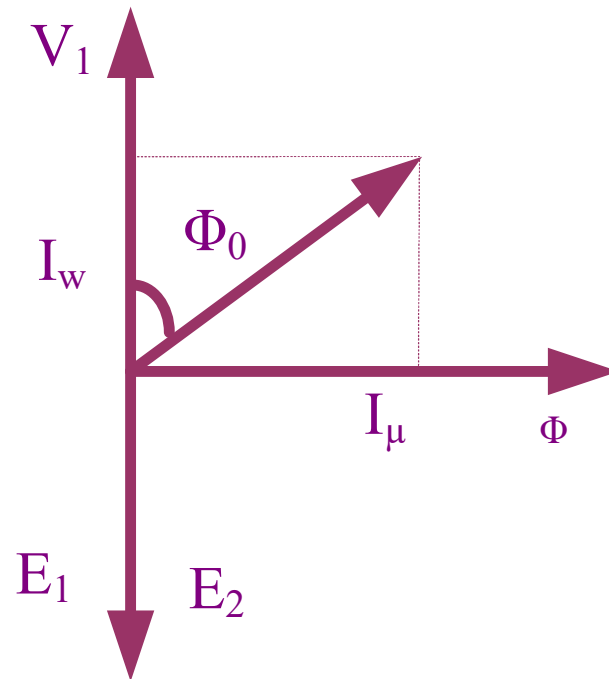
Working
Component
 $(I_w = I_0 \cos \Phi_0)$

Magnetizing
Component
 $(I_\mu = I_0 \sin \Phi_0)$

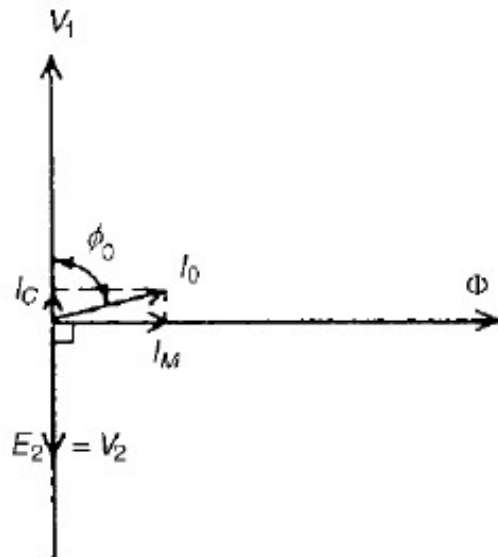
Phasor-diagram at no-load

Important Points-

- The no-load primary current is 1 to 1.5% of full load current.
- Since the permeability of the core varies with the instantaneous value of exciting current, the exciting current is not fully sinusoidal.
- No-load copper loss is negligible. Hence no-load input is practically equal to iron loss in the transformer.
- The no-load power factor is poor and lagging.



Phasor Diagram of Practical Transformer



A no-load phasor diagram for a practical transformer is shown in Figure . If current flows then losses will occur. When losses are considered then the no-load current I_0 is the phasor sum of two components — (i) I_M , **the magnetizing component**, in phase with the flux, and (ii) I_C , **the core loss component** (supplying the hysteresis and eddy current losses). From Figure

No-load current, $I_0 = \sqrt{I_M^2 + I_C^2}$, where $I_M = I_0 \sin \phi_0$ and

$$I_C = I_0 \cos \phi_0$$

Power factor on no-load = $\cos \phi_0 = \frac{I_C}{I_0}$

The total core losses (i.e. iron losses) = $V_1 I_0 \cos \phi_0$

Ex. 2 The no-load current of a 4400/440 V, single phase, 50 Hz transformer is 0.04. It consumes power 80 W at no-load when supply is given to LV side and HV side is kept open. Calculate the iron loss component and magnetizing of current.

Sol.

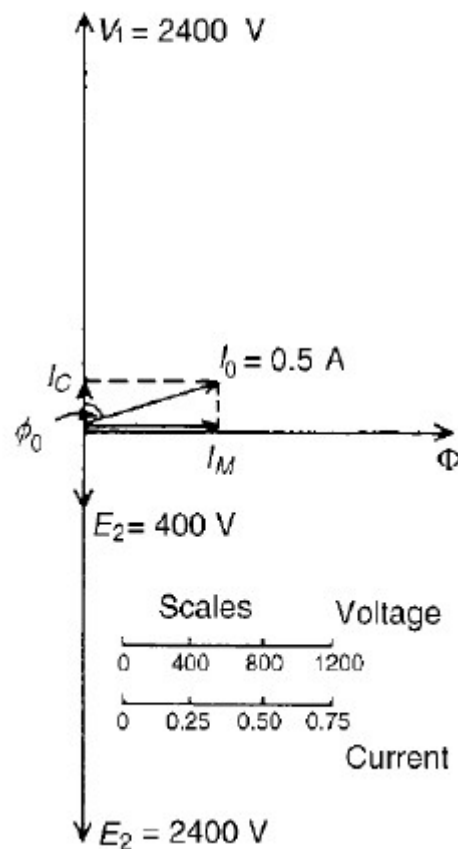
$$\cos \phi_0 = \frac{W_0}{VI_0} = \frac{80}{4400 * 0.04} = 0.454$$

$$I_w = I_0 \cos \phi_0 = 0.04 * 0.454 = 0.0187 A$$

$$\sin \phi_0 = \sqrt{1 - \cos^2 \phi_0} = 0.891$$

$$\Rightarrow I_m = I_0 \sin \phi_0 = 0.04 * 0.891 = 0.0356 A$$

Problem A 2400 V/400 V single-phase transformer takes a no-load current of 0.5 A and the core loss is 400 W. Determine the values of the magnetizing and core loss components of the no-load current. Draw to scale the no-load phasor diagram for the transformer.



$$V_1 = 2400 \text{ V}, V_2 = 400 \text{ V}, I_0 = 0.5 \text{ A}$$

$$\text{Core loss (i.e. iron loss)} = 400 = V_1 I_0 \cos \phi_0$$

$$\text{i.e. } 400 = (2400)(0.5) \cos \phi_0$$

$$\text{Hence } \cos \phi_0 = \frac{400}{(2400)(0.5)} = 0.3333$$

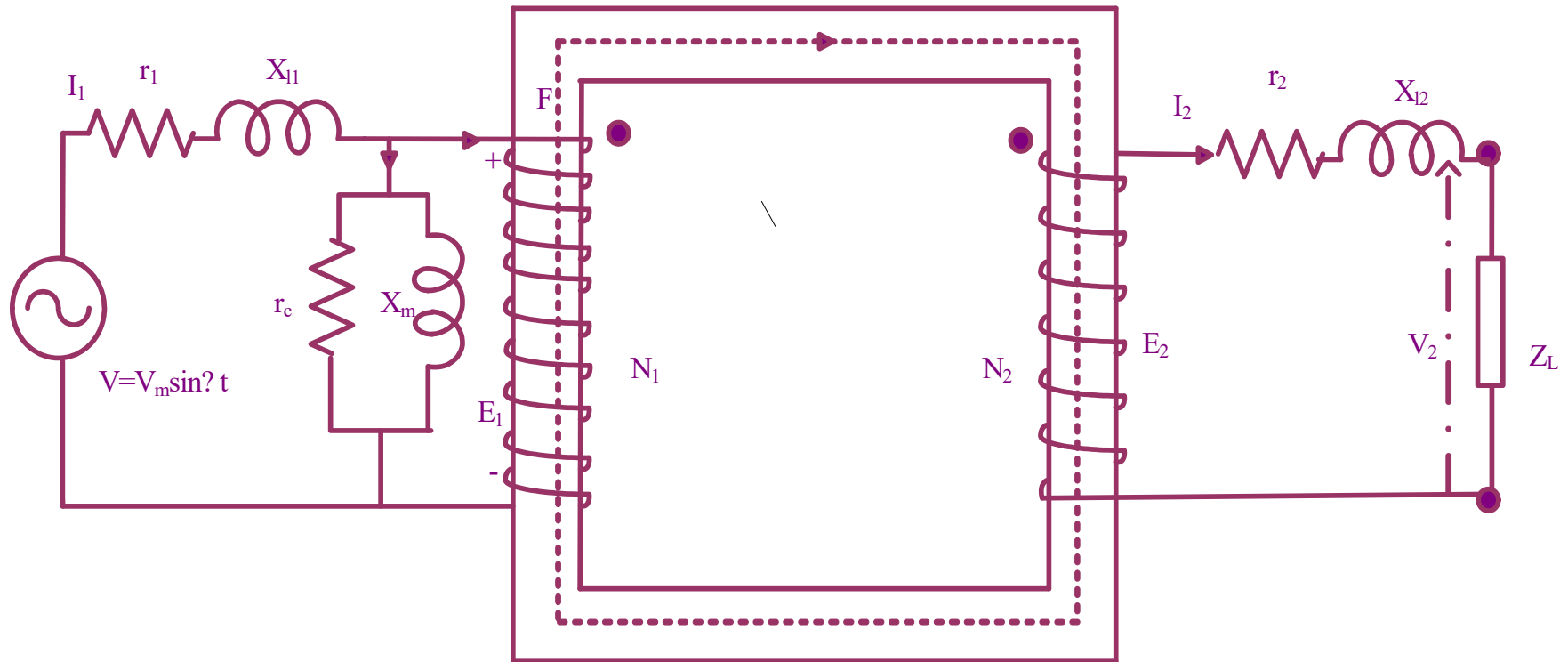
$$\phi_0 = \arccos 0.3333 = 70.53^\circ$$

The no-load phasor diagram is shown in Figure

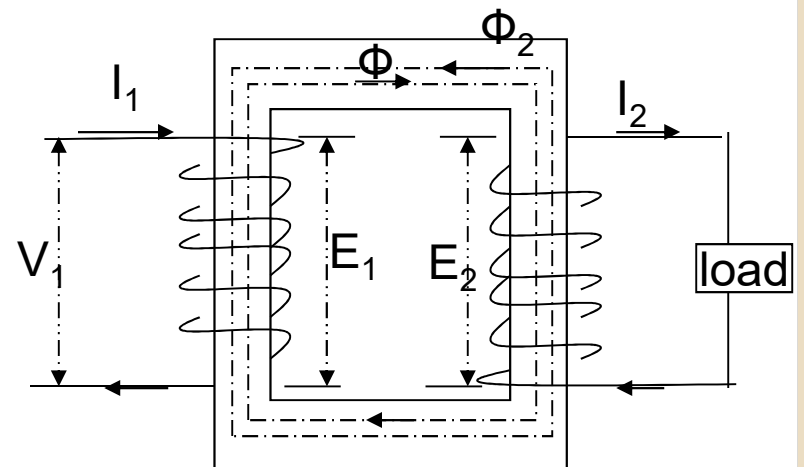
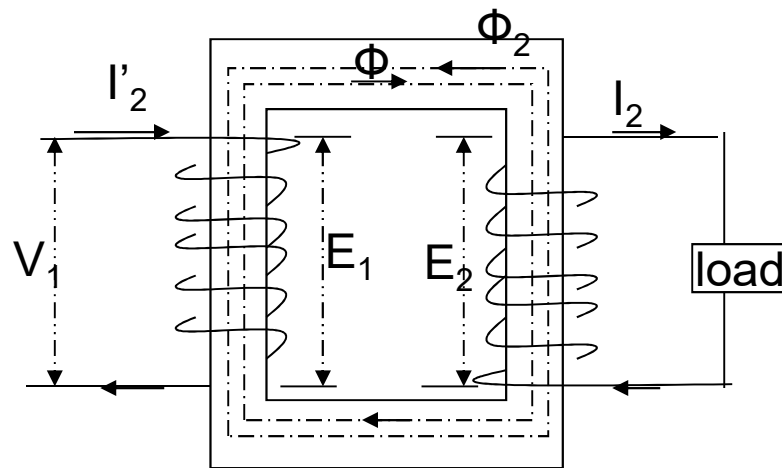
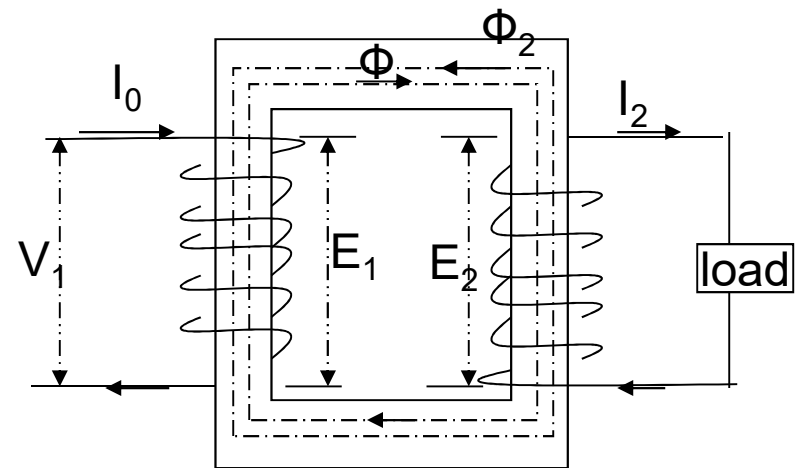
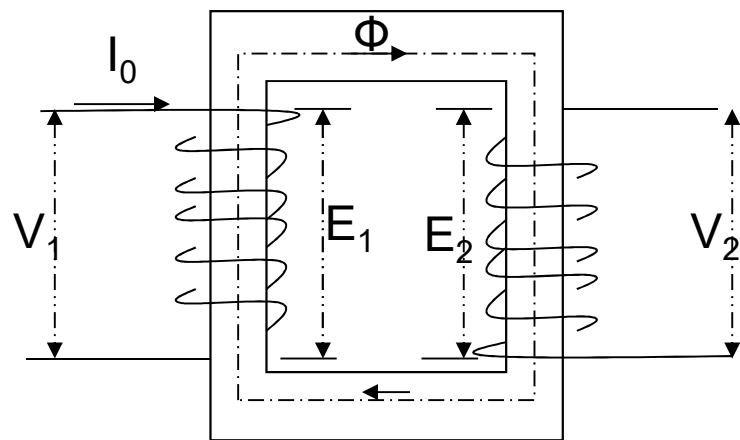
$$\text{Magnetizing component, } I_M = I_0 \sin \phi_0 = 0.5 \sin 70.53^\circ = \mathbf{0.471 \text{ A}}$$

$$\text{Core loss component, } I_C = I_0 \cos \phi_0 = 0.5 \cos 70.53^\circ = \mathbf{0.167 \text{ A}}$$

Practical Single-Phase Transformer on Load



Physical Arrangement of Practical Transformer
on Load



Practical Transformer at load

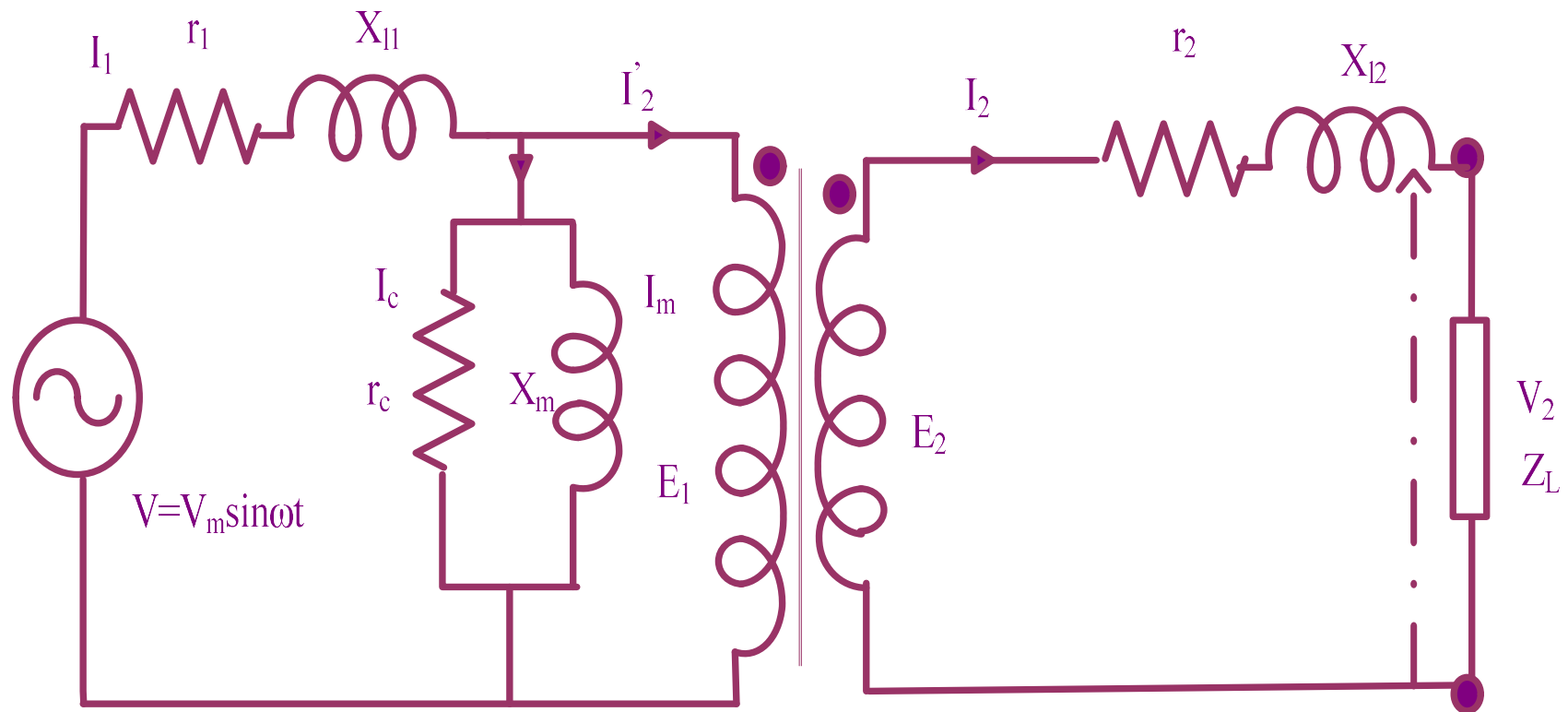
- As Shown in previous discussion that flux Φ is set up in the core during no-load operation.
- When the secondary is loaded, the secondary current will set up its own flux (Φ_2) which opposes Φ . The resultant flux becomes ($\Phi_2 - \Phi$).
- The value of e_1 will decrease because magnitude of Φ decreases.
- The v_1 becomes greater than e_1 and the primary windings draws more current from the source.
- Let the additional current drawn from the primary be I'_2 .

This current I'_2 will set up its own flux (Φ'_2) in the same direction of Φ and will oppose Φ_2 as shown in Fig. in next slide the resultant flux will be Φ at any load conditions if and only if-

$$\Phi'_2 = \Phi_2$$

$$N_1 I'_2 = N_2 I_2$$

$$I'_2 = (N_2/N_1) I_2 = I_2/a$$



Equivalent Circuit of Practical Transformer

Practical Transformer at load

- Finite μ of the magnetic core makes a few lines of flux take to a path through the air.
- Thus these flux lines do not link the secondary winding. It is called as **leakage flux**.
- As the path of the leakage flux is mainly through the air the flux produced varies linearly with the primary current I_1 .
- Even a large value of the current produces a small value of flux. This flux produces a voltage drop opposing its cause, which is the current I_1 .

$$x_{l1} = 2f l_{l1} \text{ ohm}$$

- A similar effect takes place on the secondary side when the transformer is loaded. The secondary leakage reactance jx_{l_2} arising out of the secondary leakage inductance l_{l_2} is given by

$$x_{l_2} = 2\pi f l_{l_2} \text{ ohm}$$

- The primary and secondary windings are wound with copper (sometimes aluminum in small transformers) conductors, thus the windings have a finite resistance (though small).
- These are represented by r_1 and r_2 respectively on primary and secondary side. A practical transformer have these imperfections (taken out and represented explicitly in the electric circuits) like ideal transformer of turns ratio $N_1 : N_2$ (voltage ratio $E_1 : E_2$).

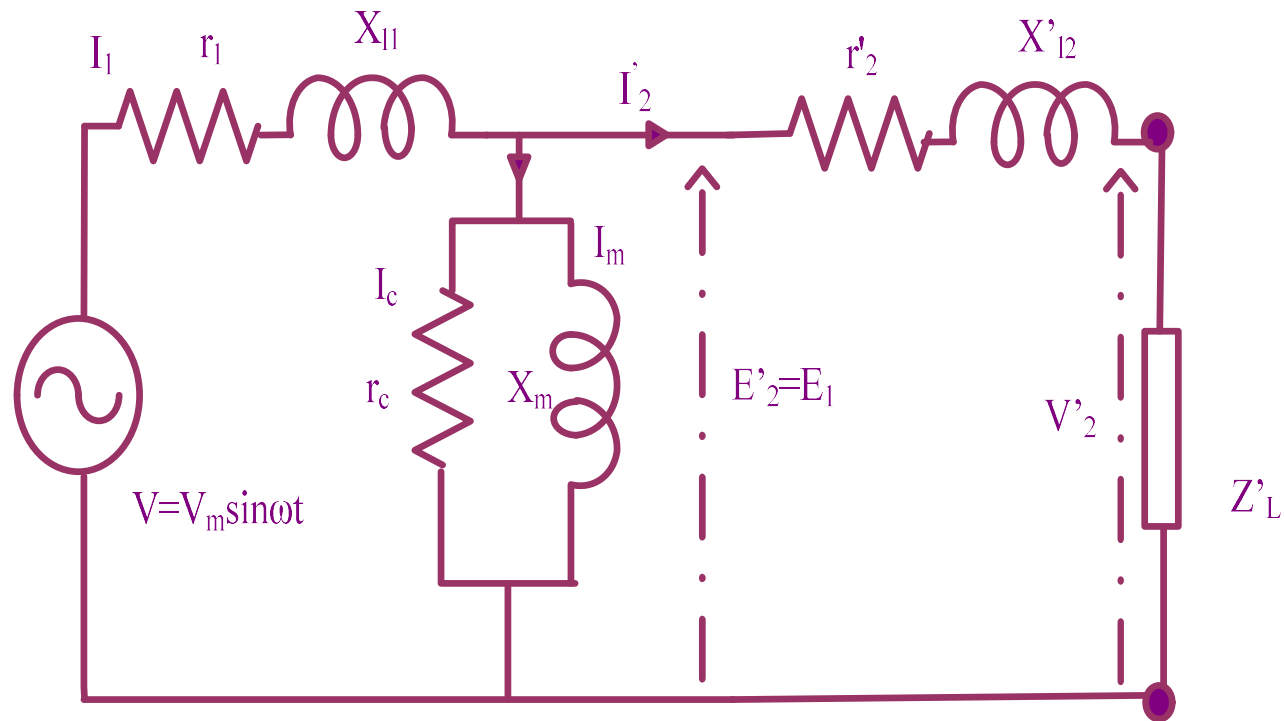
- I'_2 in the circuit represents the primary current component that is required to flow from the mains in the primary N_1 turns to neutralize the demagnetizing secondary current I_2 due to the load in the secondary turns. The total primary current

$$\text{Vectorically } \bar{I}_1 = I'_2 + I_0$$

$$\text{Here } I'_2 N_1 = I_2 N_2$$

$$\text{Thus } \bar{I}_1 = \frac{I_2 N_2}{N_1} + I_0$$

- By solving this circuit for any load impedance Z_L one can find out the performance of the loaded transformer.



Exact Equivalent Circuit Referred to Primary Side of Transformer

- If the turns ratio could be made unity by some transformation the circuit becomes very simple to use.
- This is done by replacing the secondary by a ‘hypothetical’ secondary having N_1 turns which is ‘equivalent’ to the physical secondary.
- The equivalence implies that the ampere turns, active and reactive power associated with both the circuits must be the same. Then there is no change as far as their effect on the primary is considered.

$$V_2' = a V_2$$

$$I_2' = \frac{I_2}{a}$$

$$r_{12}' = a^2 r_{12}$$

$$X_{l2}' = a^2 X_{l2}$$

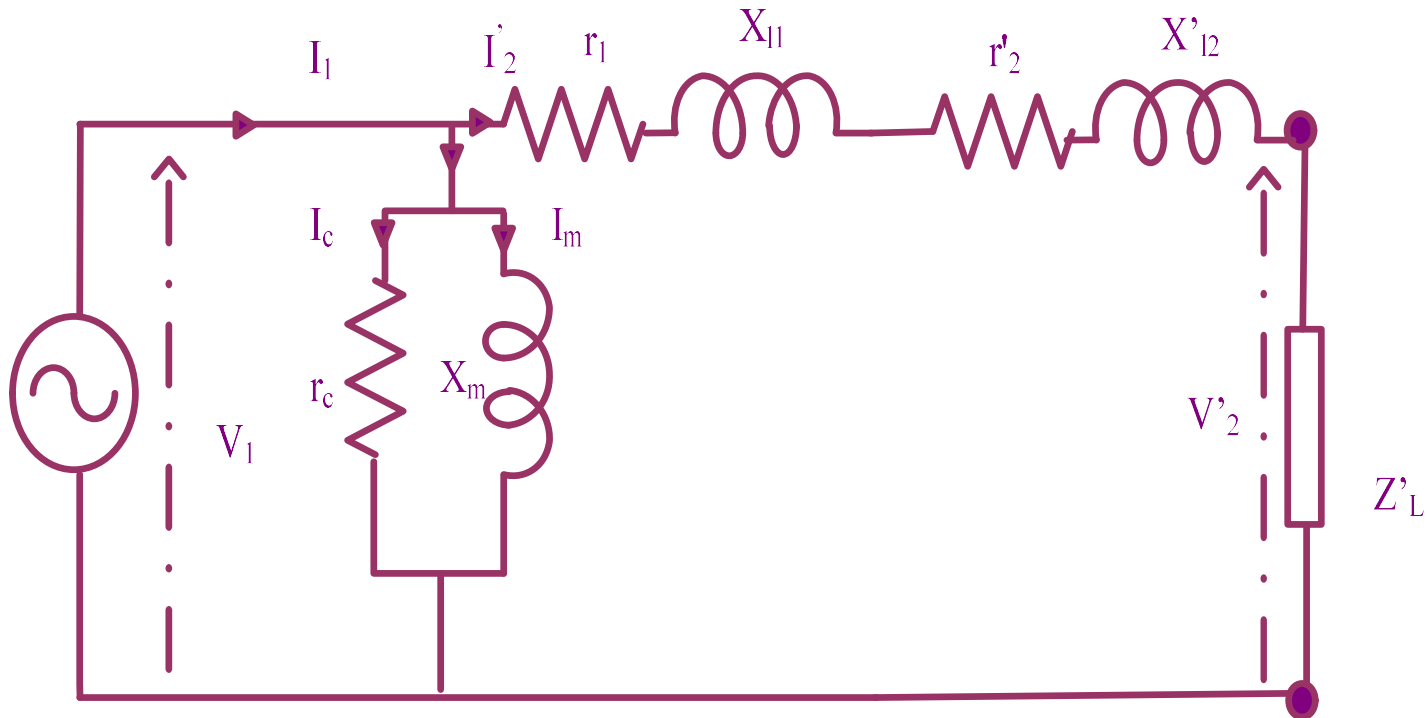
$$Z_L' = a^2 Z_L, \text{ where } a = \frac{N_1}{N_2}$$

Thus equivalent reactance referred to primary is

$$X_{01} = X_{11} + a^2 X_{12}$$

And equivalent resistance referred to primary is

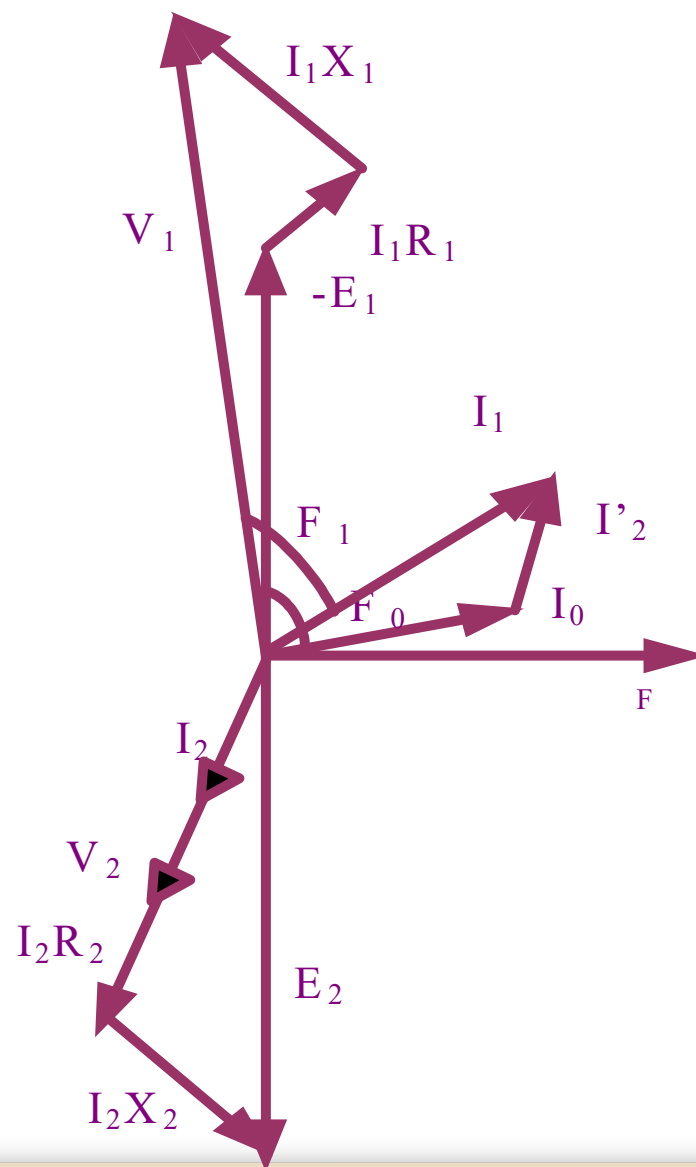
$$R_{01} = R_{11} + a^2 R_{12}$$



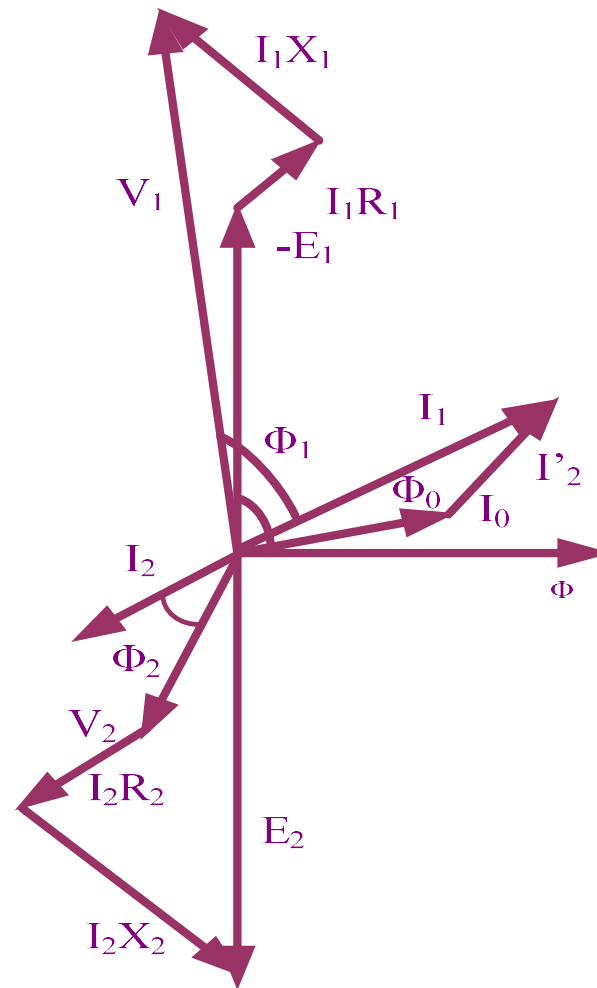
Approx. Equivalent Circuit Referred to Primary Side of Transformer

As the no-load current is less than 5% of the load current a simplified circuit known as ‘approximate’ equivalent circuit (see Fig. is usually used.

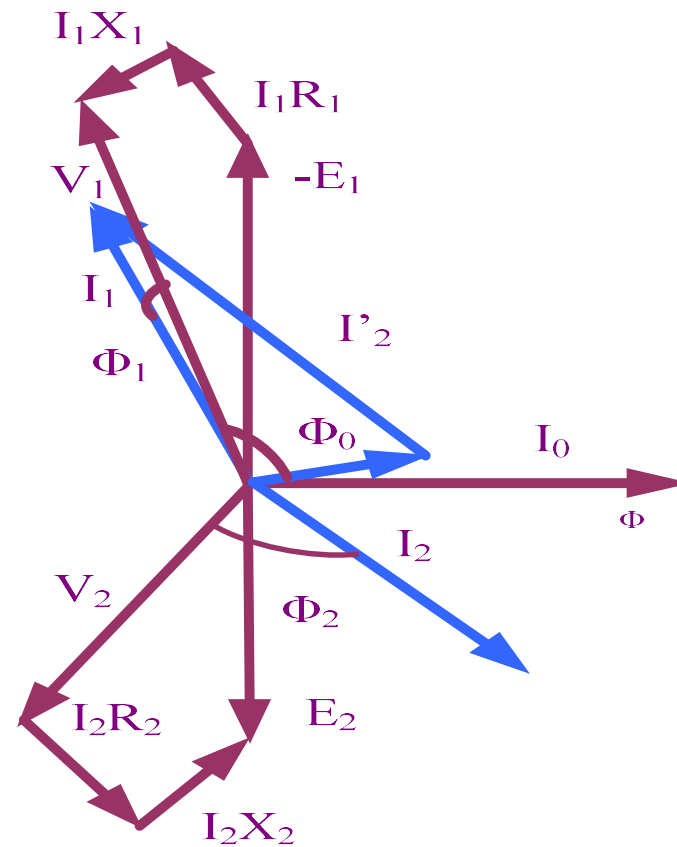
Practical Transformer at unity-power factor load



Practical Transformer at lagging-power factor load



Practical Transformer at leading-power factor load



Ex.3 A 30 kVA, 3000/300V, 50 Hz, single phase transformer has the following winding resistances and leakage reactance's:

$$R_1=2.5 \, \Omega, R_2=0.018 \, \Omega, X_1=3.8 \, \Omega, X_2=0.52 \, \Omega$$

Calculate the following:

- (1) Equivalent resistance, leakage reactance, and impedances referred to high voltage side.
- (2) Equivalent resistance, leakage reactance, and impedances referred to low voltage side.
- (3) Total Cu-loss of the transformer at full load condition

$$\text{Turn ratio}(a) = \frac{3000}{300} = 10$$

(1) *Equivalent* resistance referred to hv side:

$$R_{01} = R_{11} + a^2 R_{12} = 2.5 + (10)^2 * 0.018 = 4.3\Omega$$

Equivalent leakage reactance referred to hv side:

$$X_{01} = X_{11} + a^2 X_{12} = 3.8 + (10)^2 * 0.052 = 9\Omega$$

Equivalent impedance referred to hv side:

$$Z_{01} = \sqrt{R_{01}^2 + X_{01}^2} = \sqrt{4.3^2 + 9^2} = 9.974\Omega$$

(2) *Equivalent* resistance referred to lv side:

$$R_{02} = \frac{R_{11}}{a^2} + R_{12} = \frac{2.5}{10^2} + 0.018 = 0.043\Omega$$

Equivalent leakage reactance referred to lv side:

$$X_{02} = \frac{X_{11}}{a^2} + X_{12} = \frac{3.8}{10^2} + 0.052 = 0.09\Omega$$

Equivalent impedance referred to lv side:

$$Z_{02} = \sqrt{R_{02}^2 + X_{02}^2} = \sqrt{0.043^2 + 0.09^2} = 0.09974\Omega$$

(3) *Total* Cu loss at full load:

$$\text{Primary current}(I_1) = \frac{30 * 10^3}{3000} = 10 A$$

$$\text{Secondary current}(I_2) = \frac{30 * 10^3}{300} = 100 A$$

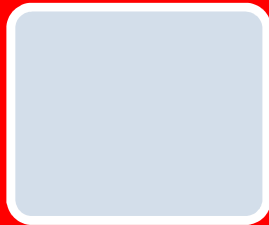
$$\text{Total cu loss} = I_1^2 R_{01} = (10)^2 * 4.3 = 430 W$$

$$\text{Also Total cu loss} = I_2^2 R_{02} = (100)^2 * 0.043 = 430 W$$

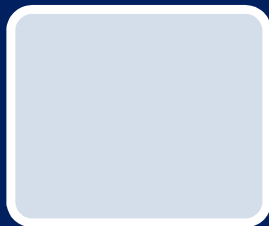
Type of Tests



Open Circuit Test

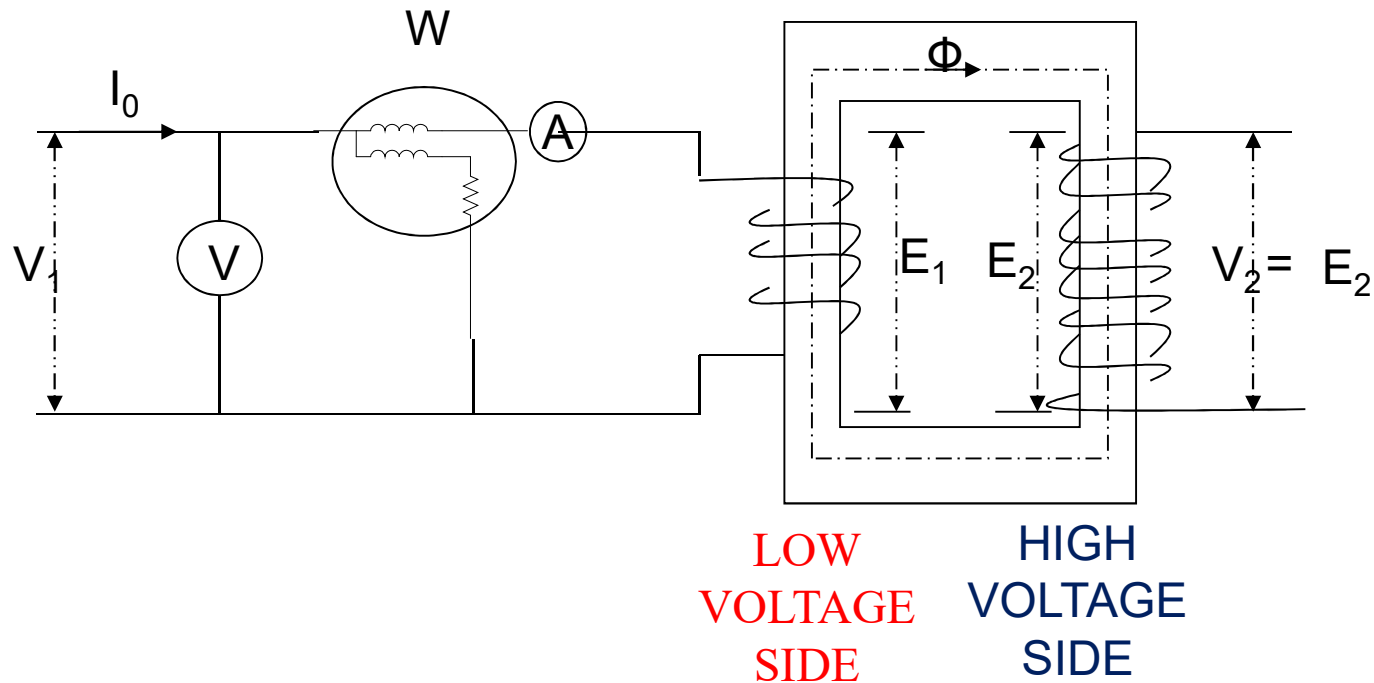


Short Circuit Test



Polarity test

Open Circuit or No-load Test



Objectives-

- To determine core loss and no-load current of the transformer.
- To determine R_0 and X_0

- The high voltage winding is generally kept open because the current in high voltage winding is less compared to that on low voltage winding.
- In low voltage side a voltmeter, an ammeter and a wattmeter are connected to measure the input voltage, no-load current and the core loss of the transformer.
- No-load current is very small, the copper loss at no-load condition is negligible.
- The wattmeter reading practically gives the iron-loss of the transformer.

Let I_0 be the reading of ammeter,

V_1 be the reading of the voltmeter and

W be the reading of the wattmeter. so we can write

$$W = V_1 I_0 \cos \phi_0$$

$$\cos \phi_0 = \frac{W}{V_1 I_0}$$

$$I_w = I_0 \cos \phi_0$$

$$I_\mu = I_0 \sin \phi_0$$

During no-load condition, the voltage drop across the primary impedance is small. Therefore we can write

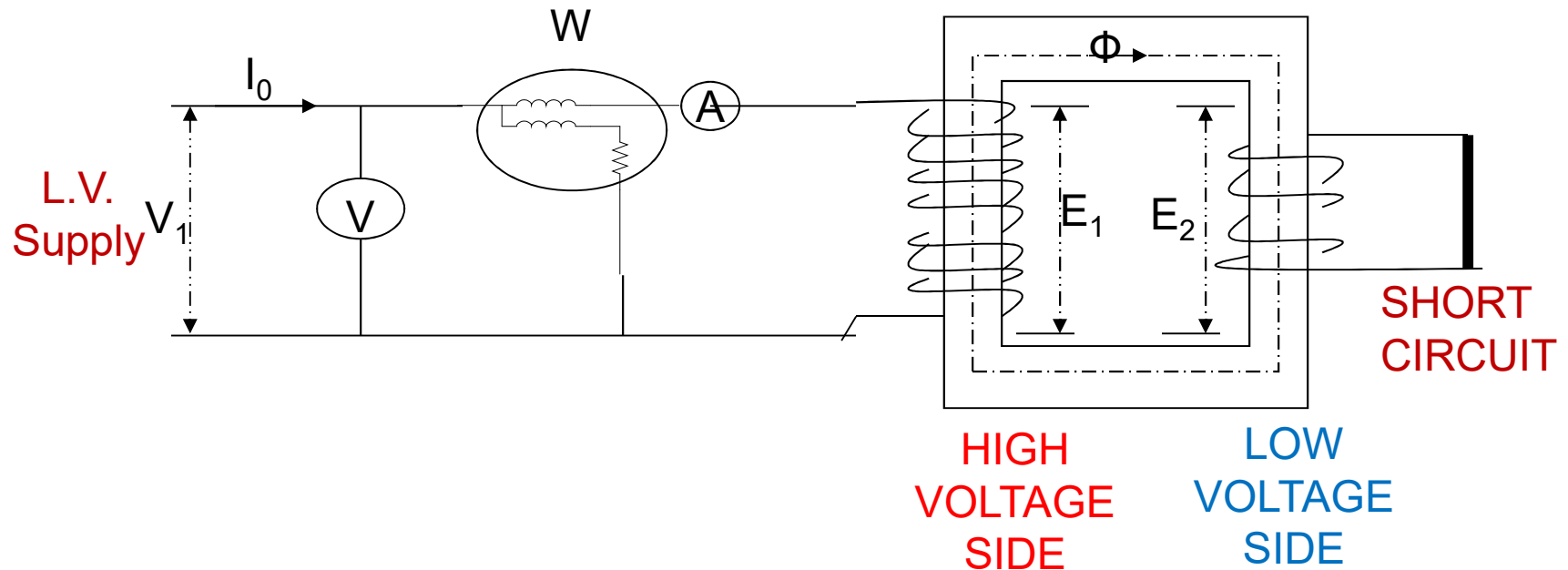
$$X_0 = \frac{V_1}{I_\mu}$$

$$R_0 = \frac{V_1}{I_w}$$

Short-circuit test

Objectives:

- To find out the equivalent impedance (Z_{01} OR Z_{02}), leakage reactance(X_{01} or X_{02}) and total resistance(R_{01} or R_{02}) of the transformers referred to the winding which the measuring instruments are placed.
- Total voltage drop in the transformer as referred to the primary/secondary and regulation of the transformer has to be found.
- Copper losses of the transformer under the full load which helps in finding the efficiency of the transformer.



- During short circuit test, five to ten percent of rated voltage is required to get the full load current both in primary and secondary.
- Instruments are placed at high voltage side.
- Low voltage side is directly short circuited by a thick conductor⁴¹

Let

P_{sc} be the reading of the wattmeter

I_{sc} be the reading of the ammeter and

V_{sc} be the reading of the voltmeter

$$Z_{01} = \frac{V_{sc}}{I_{sc}}$$

$$R_{01} = \frac{P_{sc}}{I_{sc}^2}$$

$$X_{01} = \sqrt{(Z_{01}^2 - R_{01}^2)}$$

Ex. 5 A 8kVA, 500/250V, 50 Hz, single phase transformer gave the following test data

OC test (lv side)=250V, 1.2A,80W

SC test (hv side)=25V, 12.5A,90W

Determine the Equivalent circuit referred to lv side and hv side

Sol.

OC circuit

Instruments placed on lv side and hv side is kept open.

$$V_2 = 250V, I_0 = 1.2A, W_0 = 80W$$

$$\cos \phi_0 = \frac{W_0}{V_2 I_0} = \frac{80}{250 * 1.2} = 0.267$$

$$\sin \phi_0 = \sqrt{1 - \cos^2 \phi_0} = 0.964$$

$$I_w = I_0 \cos \phi_0 = 1.2 * 0.267 = 0.3204 A$$

$$I_m = I_0 \sin \phi_0 = 1.2 * 0.964 = 1.1568 A$$

$$R_0 = \frac{V_0}{I_w} = \frac{250}{0.3204} = 780.27 \Omega$$

$$\text{and } X_0 = \frac{V_0}{I_m} = \frac{250}{1.1568} = 216.11 \Omega$$

SC test

Instruments placed on hv side and lv side is short circuited

$$V_{sc} = 25V, I_{sc} = 12.5A, P_{sc} = 90W$$

$$Z_{01} = \frac{V_{SC}}{I_{SC}} = \frac{25}{12.5} = 2\Omega$$

$$R_{01} = \frac{P_{SC}}{I_{SC}^2} = \frac{90}{(12.5)^2} = 0.576\Omega$$

$$X_{01} = \sqrt{Z_{01}^2 - R_{01}^2} = \sqrt{2^2 - 0.576^2} = 1.915\Omega$$

$$\text{Turn ratio}(a) = \frac{500}{250} = 2$$

$$R_{02} = \frac{R_{01}}{a^2} = \frac{0.576}{4} = 0.144\Omega$$

$$X_{02} = \frac{X_{01}}{a^2} = \frac{1.915}{4} = 0.4787\Omega$$

$$(I_2)_{full-load} = \frac{8 * 10^3}{250} = 32A$$

Polarity Test

Polarity test of a transformer is vital before connecting them in parallel. Otherwise with incorrect polarity, it is not possible to connect them in parallel.

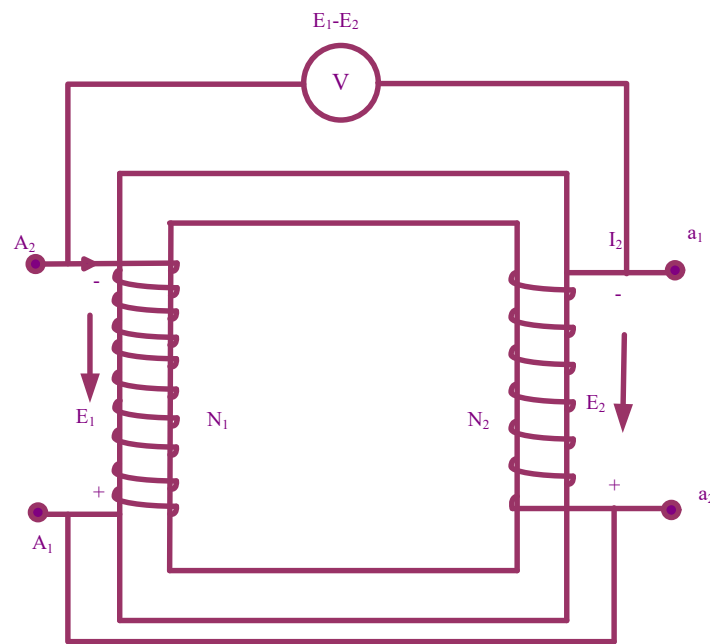


Fig. 1

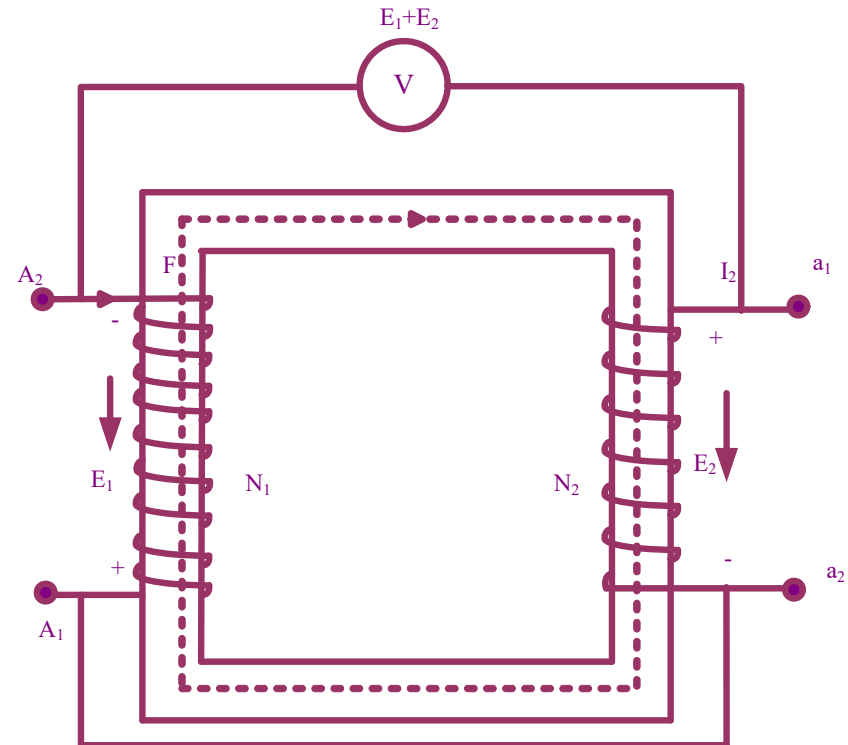


Fig. 2

Polarity Test

- The rated voltage is applied to the primary and its two terminals are marked as A_1 and A_2 respectively as shown in Fig. 1 and 2.
- The secondary winding terminals are also marked as a_1 and a_2 .
- A Voltmeter is connected across A_2 and a_2 .
- If voltmeter measures the difference of E_1 and E_2 , A_2 and a_2 are of the same polarity.
- If it measures the addition of E_1 and E_2 , A_2 and a_2 are of opposite polarity.

Losses in the transformer

- Transformer is a static device, so there is no friction and the windage losses.
- The basic losses in the transformer are
 - Core or iron losses.
 - Copper losses.

CORE OR IRON LOSSES:

There is a classification in the core losses . They are

- Hysteresis loss. $W_h = \eta B_{\max}^{1.6} f v$
- Eddy current losses. $W_e = P B_{\max}^2 f^2 t^2 v^2$

- The core flux is constant for all the loads. so the core losses are same for all loads.
- Hysteresis loss
- Eddy current loss.
- These losses can be minimized with the help of using the high silicon content (for hysteresis loss) and using the laminated core(for eddy current losses)

Copper losses:

This losses is due to the ohmic resistance in the transformer windings. the total copper losses in the transformer are

$$I_1^2 R_1 + I_2^2 R_2 = I_1^2 R_{01} = I_2^2 R_{02}$$

EFFICIENCY of the Transformer.

- Efficiency is defined as the ratio of the rated output to the rated input.
 - $\text{Efficiency} = \text{output} / \text{input}$.
- $\text{input} = \text{output} + \text{losses}$. so,
 - $\text{efficiency} = \text{output} / (\text{output} + \text{losses})$
- Losses are
 1. iron or core losses
 2. **copper losses.**
 - $\text{Efficiency} = \text{output} / (\text{output} + \text{cu losses} + \text{iron losses})$
- The efficiency in terms of input can be written as
 - $\text{Efficiency} = (\text{input} - \text{losses}) / \text{input}$
- From this
 - $\text{Efficiency} = 1 - \text{losses} / \text{input}$.

Condition For The MAXIMUM EFFICIENCY of the TRANSFORMER

$$\text{Power input} = V_1 I_1 \cos \phi_1$$

$$EFFICIENCY(\eta) = \frac{V_1 I_1 \cos \phi_1 - \text{Losses}}{V_1 I_1 \cos \phi_1}$$

$$\eta = \frac{V_1 I_1 \cos \phi_1 - I_1^2 R_{01} - W_i}{V_1 I_1 \cos \phi_1}$$

$$= 1 - \frac{I_1 R_{01}}{V_1 \cos \phi_1} - \frac{W_i}{V_1 I_1 \cos \phi_1}$$

Differentiating on both sides with respect to I_1

$$\frac{d\eta}{dI_1} = 0 - \frac{R_{01}}{V_1 \cos \phi_1} + \frac{W_i}{V_1 I_1^2 \cos \phi_1}$$

For the efficiency to be maximum

$$\frac{d\eta}{dI_1} = 0$$

$$\frac{R_{01}}{V_1 \cos \phi_1} = \frac{W_i}{V_1 I_1^2 \cos \phi_1}$$

or

$$I_1^2 R_{01} = W_i$$

So efficiency will be maximum when the constant losses is equal to the variable losses or the copper losses is equal to the iron losses.

Load Current at Maximum Efficiency

Let I_{2m} be the load current at maximum efficiency

$$\therefore I_{2m}^2 R_{02} = P_i$$

$$\text{or } I_{2m}^2 = \frac{P_i}{R_{02}}$$

If I_{2fl} be the full load current

$$I_{2m}^2 = \frac{P_i I_{2fl}^2}{I_{2fl}^2 R_{02}} = I_{2fl}^2 \frac{P_i}{P_{cuf1}}$$

$$I_{2m} = I_{2fl} \sqrt{\frac{P_i}{P_{cuf1}}}$$

Equation gives the load current in terms of full-load current at maximum efficiency

ALL-DAY EFFICIENCY of the transformer.

- Distribution transformers are used for supplying lighting and general networks and energized throughout the day.
- Their secondaries are at no-load most of the time in a day except during the hours of lighting periods.
- Core losses occurs through-out the day.
- Copper losses occurs only when they are loaded and hence less important.
- To judge the performance, all day efficiency or operational efficiency is calculated.

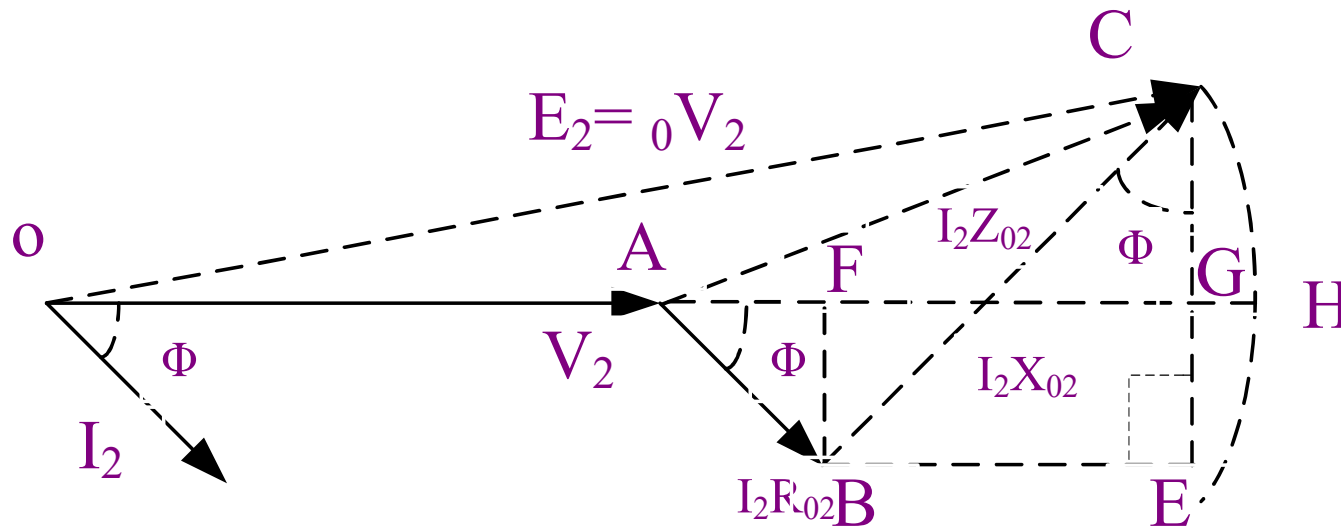
The all day efficiency is given as,

$$\eta_{all-day} = \frac{\text{output in kWH}}{\text{Input in kWH}} \text{ (for 24 hrs)}$$

The all day efficiency is always less than the commercial efficiency of a transformer.

Approximate Voltage drop of the transformer.

During no-load, induced voltages at the primary and secondary windings are equal to the applied voltage and terminal voltage at no-load respectively. If $_0V_2$ be the secondary terminal voltage at no-load, so it can be written as $E_2 = _0V_2$.



Let V_2 be the secondary terminal voltage at no-load. To calculate the approximate voltage drop of the transformer phasor diagram shown in previous slide is considered-

The approximate voltage drop

$$=AG=AF+FG =AF+BE$$

$$=I_2R_{02} \cos\Phi+I_2X_{02}\sin \Phi$$

This approximate voltage drop is due to lagging power factor.

For leading power factor, the approximate voltage drop is

$$=I_2R_{02} \cos\Phi-I_2X_{02}\sin \Phi$$

The above calculations are referred to secondary.

% voltage drop in secondary is

$$= \{ (I_2 R_{02} \cos \Phi + I_2 X_{02} \sin \Phi) / V_2 \} * 100$$

$$= V_r \cos \Phi + V_x \sin \Phi$$

Where $V_r = (I_2 R_{02} / V_2) * 100$ (resistive drop)

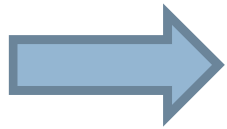
and $V_x = (I_2 X_{02} / V_2) * 100$ (reactive drop)

Exact Voltage Drop

From phasor diagram, the exact voltage drop is AH instead of AG.

From previous calculation AG has already calculated. If GH is being added to AG, the exact voltage drop can be obtained.

Considering the right angle triangle OCG. We can write



$$OC^2 = OG^2 + GC^2$$

$$\text{Or } OC^2 - OG^2 = GC^2$$

$$(OC - OG)(OC + OG) = GC^2$$

Taking $OC = OG$

$$GH = (GC^2 / 2 * OC) = \{ (I_2 X_{02} \cos \Phi - I_2 R_{02} \sin \Phi)^2 / 2 V_2 \}$$

For lagging power factor exact voltage drop is $AG + GH$

$$= (I_2 R_{02} \cos \Phi + I_2 X_{02} \sin \Phi) + \{ (I_2 X_{02} \cos \Phi - I_2 R_{02} \sin \Phi)^2 / (2 * V_2) \}$$

For leading power factor exact voltage drop is

$$= (I_2 R_{02} \cos \Phi - I_2 X_{02} \sin \Phi) + \{ (I_2 X_{02} \cos \Phi + I_2 R_{02} \sin \Phi)^2 / (2 * V_2) \}$$

Voltage regulation of a transformer

$$\text{Voltage regulation} = \frac{\text{no-load voltage} - \text{full-load voltage}}{\text{no-load voltage}}$$

recall $\frac{V_2}{V_1} = \frac{N_2}{N_1}$

Secondary voltage on no-load $V_2 = V_1 \left(\frac{N_2}{N_1} \right)$

V_2 is a secondary terminal voltage on full load

Substitute we have

If inherent voltage drop is divided by $_0V_2$ it is known as
voltage regulation down

$$\text{Voltage regulation} = \frac{{}_0V_2 - V_2}{{}_0V_2} pu$$

If inherent voltage drop is divided by V_2 it is known as
voltage regulation up

$$\text{Voltage regulation} = \frac{{}_0V_2 - V_2}{V_2} pu$$

- The secondary terminal voltage not only depend on load current but also on the power factor of the load.
- The regulation is said to full load provided V_2 is determined at

full load and at a specified power factor conditions.

- V_2 drops more and more with increasing load current.
- For **lagging power factor load**, $V_2 < E_2$, the voltage regulation is positive.
- For **leading power factor load**, $V_2 > E_2$, the voltage regulation is negative.

Calculation of Voltage Regulation

$$\% \mu = \frac{{}_0V_2 - V_2}{{}_0V_2} \times 100 = \frac{I_2(R_{02} \cos \phi \pm X_{02} \sin \phi)}{{}_0V_2} \times 100$$

Positive sign for lagging power factor and negative sign for leading power factor

Zero Voltage Regulation

For zero voltage regulation, we can write

$$V_1 - V_2 = 0$$

$$I_2(R_{02}\cos\Phi - X_{02}\sin\Phi) = 0$$

$$\tan\Phi = (R_{02}/X_{02})$$

The above equation gives the leading power factor at which voltage regulation becomes zero.

Maximum Voltage Regulation

Maximum voltage regulation can be obtained for lagging power factor. For maximum voltage regulation, we can determine

$$\frac{d}{d\phi} \left[\frac{I_2 (R_{02} \cos \phi + X_{02} \sin \phi)}{{}_0V_2} \right] = 0$$

$$\text{or } -R_{02} \sin \phi + X_{02} \cos \phi = 0$$

$$\text{so } \tan \phi = \frac{X_{02}}{R_{02}}$$

$$\text{and } \cos \phi = \cos \left\{ \tan^{-1} \left(\frac{X_{02}}{R_{02}} \right) \right\}$$

Ex.8 A single phase 10kVA, 2000/200 V, transformer has the following test results:

OC test (lv side): 200V, 0.8A, 60W

SC test (hv side): 40V, 4A, 70W

Calculate the following

(a) The efficiency of the transformer at half load and 0.8 power factor lagging.

(b) the load kVA at which maximum efficiency occurs and also the maximum efficiency at 0.8 power factor lagging

(c) The voltage regulation and 0.8 power factor leading on full load condition.

From OC test, iron loss of the transformer=60W

$$\text{Full load primary current}(I_1)=\frac{10*10^3}{2000}=5A$$

from SC test

$$Z_{01} = \frac{40}{4} = 10\Omega, R_{01} = \frac{70}{4^2} = 4.375\Omega,$$

$$X_{01} = \sqrt{Z_{01}^2 - R_{01}^2} = \sqrt{10^2 - 4.375^2} = 8.99\Omega$$

$$\text{Again } R_{02} = \frac{R_{01}}{a^2} = \frac{4.375}{10^2} = 0.04375\Omega$$

$$\text{and } X_{02} = \frac{X_{01}}{a^2} = \frac{8.99}{10^2} = 0.089\Omega$$

Approximate voltage drop at secondary

$$\partial V = I_2 (R_{02} \cos \phi + X_{02} \sin \phi)$$

$$I_2 \text{ at half-load} = \frac{5 \times 10}{2} = 25A$$

$$\partial V = 25 * (0.04375 * .8 + 0.0899 * 0.6) = 2.2235V$$

$$V_2' = 200 - 2.2235 = 197.7765V$$

$$\text{output power} = 197.7765 * 25 * 0.8 = 3955.63V$$

$$\text{copper - loss at half-load} = \left(\frac{2.5}{4}\right)^2 * 70 = 27.34W$$

$$\text{Effeciency of the transformer at half-load} = \frac{\frac{1}{2} * 3955.536}{\frac{1}{2} * 3955.53 + 60 + 27.34}$$

Approximate voltage drop at secondary

$$=0.9577\text{pu}=95.77\%$$

(ii) At max. efficiency cu loss=iron loss=60W

$I_1^2 R_{01} = 60W$ where I_1 is the primary current at maximum efficiency

$$I_1 = \sqrt{\frac{60}{R_{01}}} = \sqrt{\frac{60}{4.375}} = 3.703A$$

$$\text{Load kVA at maximum efficiency} = 10 * \frac{3.703}{5} = 7.406kVA$$

(iii) Voltage regulation at 0.8pf leading

$$I_2 = 50A \text{ at full load}$$

$$\partial V = I_2 (R_{02} \cos \phi - X_{02} \sin \phi)$$

$$\partial V = 50(0.04375 * 0.8 - 0.0899 * 0.6)$$

$$=-0.947V$$

$$\begin{aligned} \text{Regulation of the transformer} &= \frac{-0.947}{200} \\ &= -4.735 * 10^{-3} pu \\ &= 0.4735\% \end{aligned}$$

Ex. 9 A single phase 12kVA, 1200/400 V, transformer has the following test results:

OC test (lv side): 1200V, 1.2A, 70W

SC test (hv side): 30V, 22A, 80W

Calculate the following

(a) Power factor of the load so that regulation is zero at full load.

Sol. $Turn\ ratio = \frac{1200}{400} = 3$

from OC test

Core loss of the transformer = 70W

from SC test (Instruments placed on lv side)

$$V_{sc} = 30V, I_{sc} = 22A, P_{sc} = 80W$$

$$Z_{02} = \frac{V_{sc}}{I_{sc}} = \frac{30}{22} = 1.364\Omega,$$

$$R_{02} = \frac{P_{sc}}{I_{sc}^2} = \frac{80}{22^2} = 0.1653\Omega,$$

$$X_{02} = \sqrt{Z_{02}^2 - R_{02}^2} = \sqrt{1.364^2 - 0.1653^2} = 1.354\Omega$$

For zero voltage regulation

$$\tan \phi = \frac{R_{02}}{X_{02}} = \frac{0.1654}{1.3453} = 0.6960$$

$$\cos \phi = 0.9926(\text{leading})$$

References

1. Stephen J. Chapman, “*Electric machinery fundamentals*” The McGraw-Hill Companies, New York 2005.
2. Arthur Eugene Fitzgerald, Charles Kingsley, Stephen D. Umans “*Electrical machinery*” Tata McGraw-Hill publishing company Limited, New Delhi, India, 2009.
3. Samarjit Gosh, “Electrical Machines,” First Indian Print, 2005, Pearson Education (Singapore) Pte. Ltd .
4. D.V. Richardson, “*Rotating Electric Machinery and Transformer Technology*, ” Prentice Hall Company, Reston, Virginia, 1978.
5. I. J. Nagrath and D.P. Kothari, “ *Electric Machines*”, TMH, New Delhi, 2004.
6. M.G. Say, “*The Performance and Design of Alternating Current Machines*,” CBS Publishers and Distributors, Third Edition, Delhi, 2000.